

			Need to show constant value, to 1 S.F., for all 3 orbits (shown)	
7	(a)		When there is no current through the 12 V battery, the potential difference across the 12 V battery and the light bulbs must be equal to 12 V. If the p.d. across each light bulb is 12 V, the current through each bulb is $12/3.0 = 4.0$ A Current through resistor R = $4.0 + 4.0 = 8.0$ A p.d. across R = $14.0 - 12.0 = 2.0$ V resistance of R = $2.0 / 8.0 = 0.25 \Omega$	B1 C1 A0
	(b)		Since the potential difference across R_1 remains at 12 V, same amount of current will flow through R_1 and no current will flow through R_2. Hence, the current through the 14 V generator and resistor R will reduce to 4.0 A Since the p.d. across R remains the same at 2.0 V, the value of R should be adjusted to a lower value (new R = $2.0 / 4.0 = 0.50 \Omega$)	B1 B1 B1
	(c)		There will be emf across the lamps even if either the generator or the battery is defective. The power from the generator can be used to charge the car's battery when the light bulbs are removed. The light bulbs will light up longer since energy is delivered from both generator and battery.	B1 B1
8	(a)	(i)	$R = 226 \Omega = 23$	B1